

Research Experiences

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I. Construction of cryogenic ion trap system for quantum computing

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Supervisor: Prof. Moonjoo Lee

1. Multi-layer ion trap chip and its characterization

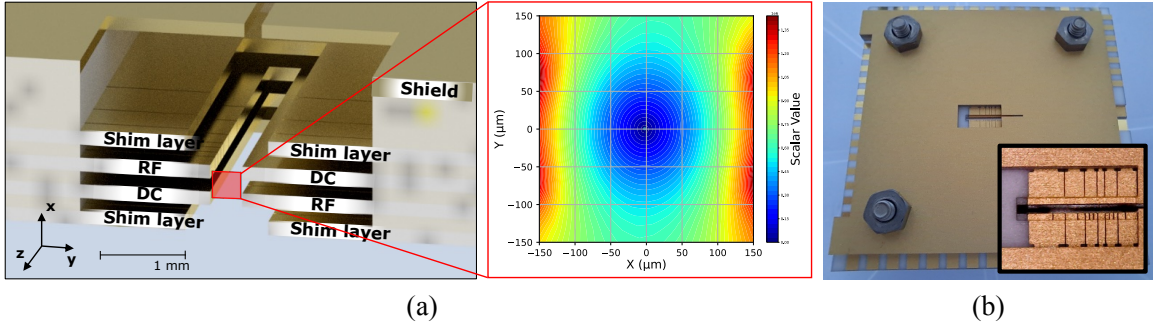


Figure 1: (a) Cross-sectional view of the designed multi-layer ion trap and simulated radial field distribution (Ansys Maxwell 3D). (b) Photograph of the fabricated multi-layer ion trap chip.

- A multi-layer ion-trap chip composed of nine stacked layers was utilized in this work. The chip includes DC/RF layers that provide the trapping potential, shim layers for principal-axis and micromotion compensation, and spacer layers that separate the electrode layers. A shield layer is positioned on the top of the chip to collimate the ablated atomic beam. The DC electrodes are segmented to enable ion shuttling, as depicted in Fig. 1(a, left).
- Simulated the designed structure with $\Omega_{\text{RF}} = 2\pi \cdot 20.26$ MHz and $V_{\text{RF}} = 75$ V using the Ansys Maxwell 3D electrostatic solver, as shown in Fig. 1(a, right). The simulations yield radial secular frequencies of about 3 MHz, in good agreement with the experimentally measured values.
- Laser-cut the alumina substrate using a picosecond laser, deposited a Ti adhesion layer and Au seed layer by sputtering, and electroplated additional Au to form the electrodes. The fabricated multi-layer chip is shown in Fig. 1(b).

2. Construction of cryogenic ion trap system

- Constructed the laser and optical system for photoionization (422 and 369 nm), Doppler cooling (397 nm), and repumping (866 and 854 nm) for ion trapping in the cryostat, as shown in Fig. 2(a).
- Constructed the cryogenic ion trap system inside the cryostat, as illustrated in Fig. 2(b). The main in-vacuum components are summarized as follows:
 - The ion trap chip was mounted and gold wire-bonded on a cryo- and vacuum-compatible PCB. Low-pass filter boards were installed on both sides of the trap and connected to the DC electrodes through ribbon cables, while the RF signal was directly fed to the trap.
 - The ion trap assembly was enclosed in an oxygen-free high thermal conductivity (OFHC) copper chamber mounted on the cold plate of the cryostat. The chamber was cooled to 4 K by a

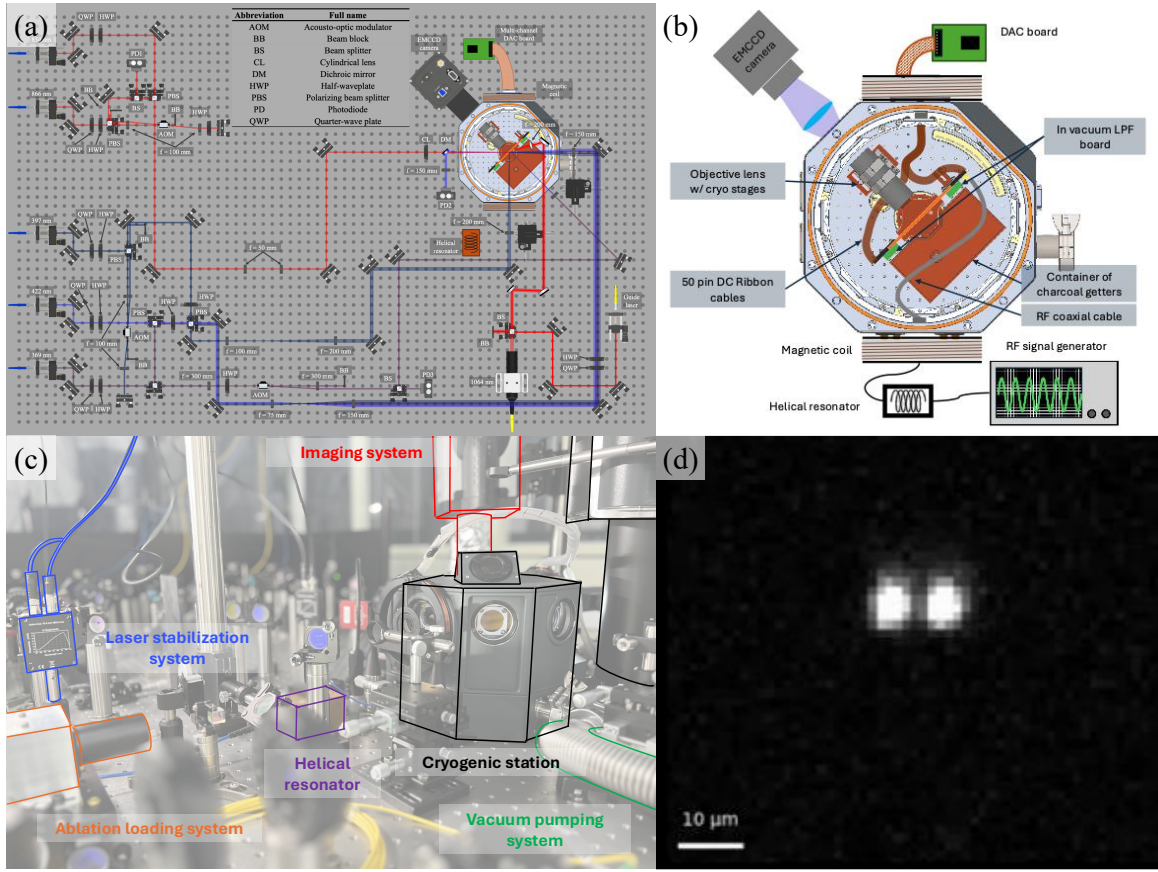


Figure 2: (a) Overall layout of the laser and optical setup for photoionization, Doppler cooling, and repumping. (b) Schematic of the cryogenic ion trap system and its main in-vacuum components. (c) Photograph of the entire cryogenic ion trap setup. (d) Fluorescence image of two trapped-ions.

pulse-tube cryocooler to provide cryopumping, and charcoal getters were incorporated to enhance cryosorption.

- A high-NA (0.38) objective lens was positioned above the trap and mounted on a cryogenic-compatible XYZ piezo translation stage for ion fluorescence imaging.
- Fig. 2(c) shows the overall cryogenic ion trap experiment, including the fluorescence imaging system with a high-gain EMCCD camera, the vacuum pumping system for maintaining the ultra-high-vacuum (UHV) environment, the helical resonator connected to the RF signal generator for stable high-Q RF delivery, the ablation loading system with a pulsed laser and calcium target, and the laser stabilization system.
- Implemented laser power stabilization and frequency locking using a PID control loop on an FPGA board together with a high-resolution frequency modulation system. These subsystems were integrated into the experimental system through a home-built PyQt-based UI.
- Demonstrated trapping of up to two ions with $(\omega_x, \omega_y) = 2\pi \cdot (2.75, 2.88)$ MHz. A representative fluorescence image of two trapped-ions is shown in Fig. 2(d).
- Presented the construction progress and preliminary experimental results of the cryogenic ion trap system at the 2024 Korean Conference on Semiconductors, where this work received an on-site outstanding poster award.

3. Investigation of light-induced charging effects on the trap electrodes

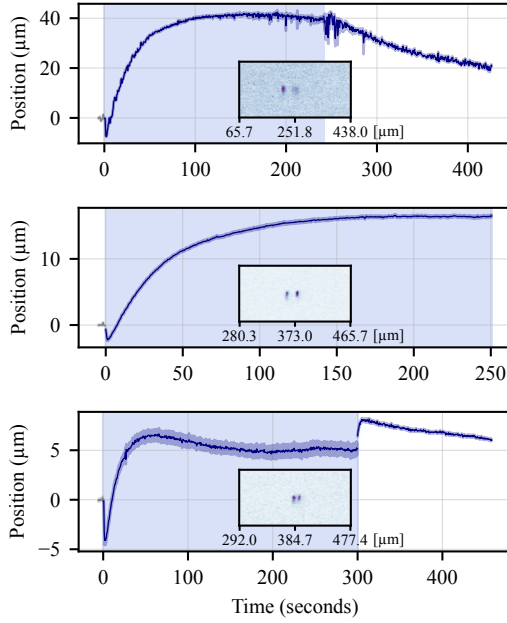


Figure 3: Axial displacement of a single trapped-ion used as an *in-situ* probe of light-induced surface potentials.

4. CNN-based detection program for trapped-ions

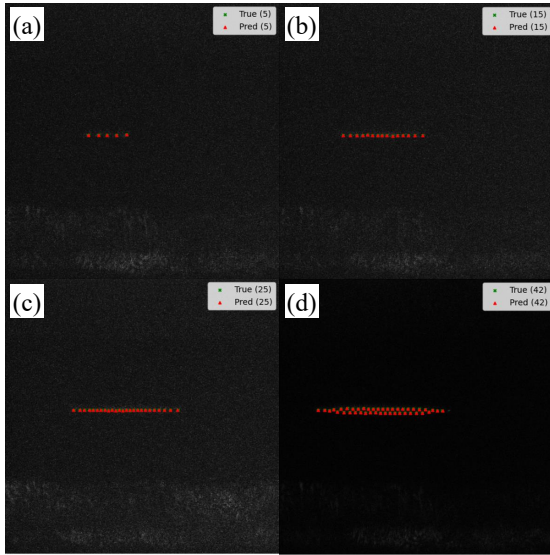


Figure 4: Example EMCCD images of trapped-ions and the corresponding detection results for (a) 5, (b) 15, (c) 25, and (d) 42 ions.

- Used a single trapped-ion as a sensitive *in-situ* probe of light-induced charging on the trap electrodes by monitoring its time-dependent axial displacement when a focused UV beam was applied to oxide patches on the electrode surface.
- Performed time-resolved measurements of the ion position during illumination and in the subsequent relaxation period, and varied beam power, exposure time, and illumination geometry to observe how these parameters qualitatively modify the charging and discharging behavior.
- Identified sign-changing charging and discharging effects on the oxide surface from the ion-image data, and compared these features with expectations from a trap-limited surface photovoltage (SPV) picture.
- Discussed how the observed photo-induced stray fields affect ion stability in our cryogenic ion-trap system.

- Developed a CNN-based detection program (Faster R-CNN) that takes EMCCD images as input and automatically identifies the number and positions of trapped-ions in real time.
- Collected and labeled experimental ion images under varying ion numbers, fluorescence levels, and noise conditions to train and validate the model, and integrated the detection results with the experimental control software for feedback loading.
- Evaluated the model on experimental EMCCD images and confirmed that the predicted ion positions and counts agree with the labeled ground truth over 42 ions, as illustrated in Fig. 4.
- Filed a Korean patent application on November 2025 under the title "Deep learning inference device for ion qubit detection, ion qubit detection system, and ion qubit detection method" based on this CNN detection framework.

II. Normally-off GaN MOS-HFET

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Supervisor: Prof. Ho-young Cha

1. Modeling of a normally-on GaN/AlGaIn HEMT

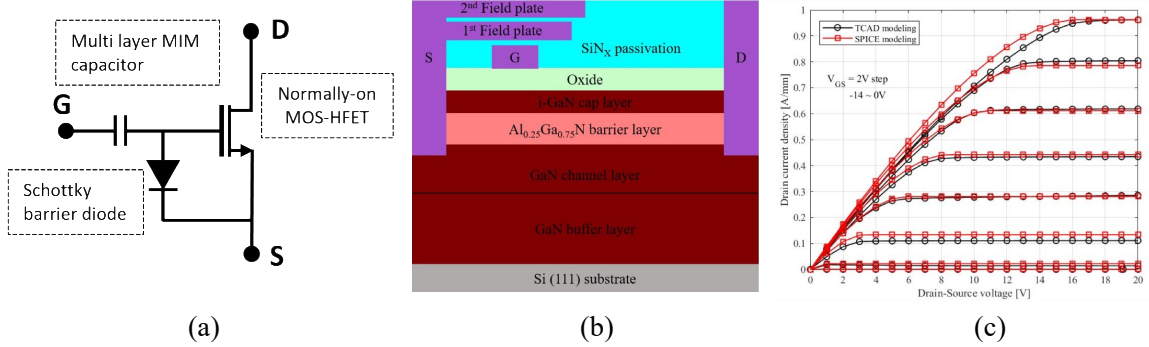


Figure 5: (a) Schematic of AlGaIn/GaN MOS-HFET integrated with a clamp circuit, (b) TCAD (Silvaco) model structure, (c) I-V characteristics of TCAD and LTspice models

- Extended previous work in which a clamp circuit was combined with an AlGaIn/GaN MOS-HFET to realize normally-off operation while targeting high switching speed and high power-conversion efficiency for high-power, high-frequency applications.
- Modelled an AlGaIn/GaN HFET in Silvaco-ATLAS by setting parameters such as unintentional doping, polarization-induced sheet charge, and surface state density so that the simulated device reproduces the measured behaviour (2DEG concentration $4.5 \times 10^{13} \text{ cm}^{-2}$, threshold voltage $\sim -14 \text{ V}$, on-resistance $\sim 12.51 \Omega \cdot \text{mm}$ at $V_G = 0 \text{ V}$).
- Extracted channel-width-dependent I-V characteristics from DC sweep simulations and capacitance characteristics from transient simulations for use in circuit design.
- Converted the TCAD-based device model into a BSIM4-compatible SPICE library so that the GaN MOS-HFET can be used in circuit-level simulations.

2. Optimization of circuit

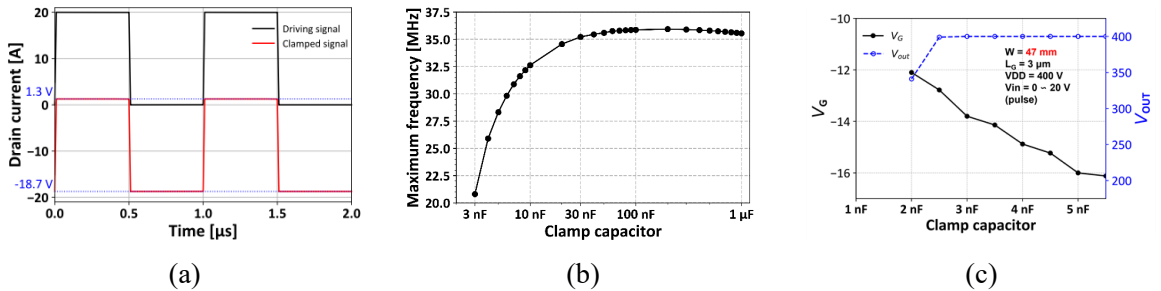


Figure 6: (a) Waveform conversion through the clamp circuit, (b) Clamp capacitance dependent maximum frequency, (c) Clamped low-side peak gate voltage and V_{ds} as a function of clamp capacitance

- Implemented normally-off operation of the D-mode GaN MOS-HFET in LTspice by adding a clamp circuit composed of a GaN Schottky barrier diode and a multilayer metal-insulator-metal capacitor, which shifts the input gate signal by about -18.7 V in peak-to-peak amplitude and enables

enhancement-mode operation.

- Chose the clamp capacitance and gate capacitance to achieve high switching speed, and verified that with a clamp capacitance of 30 nF and a channel width of 47 mm the device supports switching frequencies up to 35 MHz.

3. DC-DC boost converter circuit

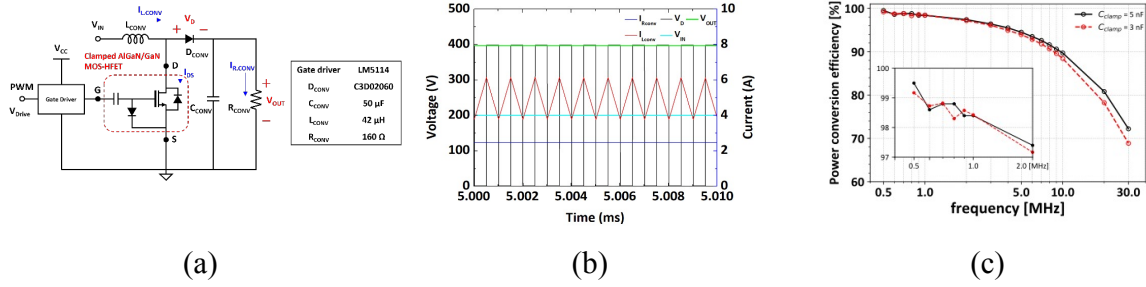


Figure 7: (a) Schematic of DC-DC boost converter, (b) switching waveforms of the boost converter, and (c) Power conversion efficiency versus frequency as a function of clamping capacitance with the device width of 47 mm.

- Verified the operation of the normally-off GaN MOS-HFET in a DC-DC boost converter and showed that, for clamp capacitances larger than 5 nF, the circuit achieves a power-conversion efficiency of up to 99.5% at a switching frequency of 1 MHz.
- Presented the results at the 2022 Conference on Semiconductor Engineering and at the 2023 Korean Conference on Semiconductors, where I received an undergraduate poster presentation award for this work.